

From learning roadmap to career evidence system

Current product inventory, business model, technical architecture, growth recommendations and the missing capabilities required to become a category leader.

18	28	252	9
career paths	skill areas	curated topics	sector lenses

The opportunity

DataPath can occupy a useful space between course libraries, static roadmaps and interview-practice tools. Its strongest proposition is not more content. It is a transparent system that tells a learner what to learn, why it matters in a target industry, how to prove it and what to revisit when evidence is weak.

Core promise

Choose a role, sector, starting point and time budget. DataPath creates a curated gap plan and connects every lesson to practice, assessment and career evidence.

Current advantage

Broad data and AI role coverage, sector context, technical quiz hierarchy, English-first guidance, evidence capture and interview/CV preparation in one product.

Strategic risk

The experience still depends heavily on reading and self-report. Without runnable practice, richer content and social proof, learners may understand the plan but lack the pull to return daily.

Recommended product thesis

Build the most credible career-readiness record for data professionals in emerging markets: curated curricula, sector fluency, technical practice, portfolio proof and interview performance in one inspectable ledger.

The product should optimise for meaningful return behaviour: finish one useful session, leave evidence, see the next clear action and feel progress toward a real job outcome.

What DataPath currently provides

The current build is already a connected platform rather than a single roadmap page.

Personalised onboarding

Role discovery, 18 career choices, experience, optional interests, self-assessment, budget preference, completion period and weekly availability.

Sector intelligence

Nine lenses covering banking, finance, marketing, healthcare, retail, SaaS, telecom, government and cross-industry fundamentals.

Curated knowledge model

28 skills and 252 levelled topics with stable IDs, prerequisites, estimates, official resources and topic-specific learning guidance.

Assessment hierarchy

Five-question lesson quizzes, ten-question level checks, twenty-question skill exams and cumulative reviews with weak-topic remediation.

Evidence system

Lesson notes, project records, skill certification state and an exportable Proof Ledger based on demonstrated work rather than decorative XP.

Career execution

Portfolio brief, interview studio, behavioural and technical cases, CV guidance, job checklist and application tracker.

Resources and tools

Free/paid resource preference, official documentation, YouTube discovery and practical explanations for required software.

Account and privacy foundation

Guest persistence, local accounts when configured, signed sessions, owner-scoped state, import/export and bounded AI consent.

The learner journey

A strong session should always answer four questions: What should I do now? Why does it matter? How do I practise it? What evidence proves I can do it?

01	Orient	Select role and sector; understand real work and fit.
02	Diagnose	Combine self-assessment with optional technical checks.
03	Learn	Read a focused explanation, objectives and official guidance.
04	Practise	Complete a hands-on task and record an edge case.
05	Validate	Pass topic, level, skill and cumulative checks.
06	Apply	Turn knowledge into a sector-relevant portfolio case.
07	Prepare	Practise technical explanation, interviews, CV and applications.
08	Prove	Export an evidence ledger showing exactly what is demonstrated.

Recent UX improvement

The dashboard hero now contains a functional next-mission console instead of decorative artwork. The console names the next skill and topic, shows estimated effort, path progress and a direct focused-session action.

Business model

The product is completely free today. Monetisation should arrive only after core learning outcomes and retention are proven.

Primary customer

Students, career switchers and working professionals pursuing data, analytics, engineering, governance and AI roles, initially with strong relevance to Egypt and the wider MENA market.

Phase	Offer	Revenue logic	Guardrail
Now	Free complete learning and career workspace	Build trust, outcomes and behavioural data	No artificial feature locks
Consumer Plus	Runnable labs, richer AI feedback, mentor review, advanced mock interviews	Monthly or annual subscription	Core curriculum and proof stay free
Institutions	Cohorts, instructor view, assignments, readiness analytics	University and training-centre licences	Learner consent and transparent metrics
Employers	Private talent challenges and verified evidence profiles	Challenge hosting or qualified-introduction fee	No hidden ranking or automated hiring decision

Channels

- Search and YouTube content built around specific job questions and sector cases.
- University career centres, technical communities and employer-sponsored challenges.
- Public portfolio profiles that learners voluntarily share on CVs and professional networks.
- Referral loops through useful project templates, interview packs and proof exports.

Market position and differentiation

Adjacent products typically excel in one layer: courses, roadmaps, coding practice, projects or interviews. DataPath should connect the layers around evidence and local relevance.

Category	Typical strength	DataPath opportunity
Course libraries	Depth and interactive lessons	Role/sector context and an evidence record across providers
Static roadmaps	Clear topic sequence	Assessment, remediation, projects and career workflow
Coding practice	Immediate technical feedback	Add execution while retaining job and sector narrative
Interview platforms	Question volume and company patterns	Connect answers to actual curriculum gaps
Portfolio platforms	Public presentation	Show the learning and validation behind every artifact

Defensible advantages to build

- A high-quality role-skill-topic graph with stable assessment and evidence semantics.
- Sector overlays authored with practitioners, starting with banking and financial services.
- A longitudinal learner record that explains readiness instead of producing a mysterious score.
- English-first content with regional examples, salary-free career realism and practical employer expectations.

Technical architecture

The current implementation is a React and TypeScript application with deterministic shared learning logic and an Express/tRPC backend.

CLIENT	SHARED DOMAIN	SERVER	PERSISTENCE
React 19 Vite React Query/tRPC Responsive CSS Local guest state	Zod state schema 18 roles 28 skills / 252 topics Planner and prerequisites Quiz and evidence logic	Express tRPC routers Authentication AI boundary Validation and rate limits	Drizzle ORM MySQL when configured Owner-scoped records Optimistic revisions JSON import/export

Planning behavior

- Requirements combine the selected role with optional skill interests and transitive prerequisites.
- Self-assessed known levels are excluded from gaps but are not treated as certification.
- A failed foundation check can reopen earlier topics; quiz failures add weak topics to review.
- Time estimates affect scheduling and feasibility, never remove required curriculum.
- AI can personalise feedback against bounded topic IDs but cannot invent curriculum or mark work complete.

Security, privacy and operational readiness

The project has a useful security foundation, but production deployment still requires configured infrastructure and operational controls.

Identity

Local password flow uses scrypt when the database is configured. Sessions use signed HttpOnly cookies. OAuth integration remains provider-specific.

Ownership

Learning records are user-scoped. Revision checks reject stale writes instead of silently overwriting newer progress.

Input safety

Zod bounds state fields, imports and AI inputs. AI output topic IDs are checked against the supplied curriculum subset.

Browser safety

CSP, frame denial, no-sniff, referrer controls, HSTS in production and same-origin write checks reduce common web risks.

Data choice

Guest data remains in browser storage. AI transmission requires authentication, explicit consent and a configured provider.

Current limits

No live production database, generic identity provider, distributed rate limiting, observability stack or penetration test is included yet.

Production checklist

- Managed MySQL with backups, tested restoration and schema migration process.
- Generic OIDC or a well-supported identity service with email verification and recovery.
- Centralised logs, error reporting, metrics, uptime checks and audit events.
- Distributed abuse controls, secrets management, privacy policy and retention controls.
- Accessibility review, threat model, dependency scanning and independent security test.

Make the experience worth returning to

Retention should come from clarity, useful practice and visible evidence - not fake streak pressure or noisy gamification.

One compelling mission

Open with the next meaningful task, estimated effort and a direct start action. The current hero now follows this pattern.

A real work surface

Let learners run SQL and Python, inspect results and receive deterministic tests inside the lesson.

Visible payoff

Show which interview capability, project artifact or proof requirement each topic unlocks.

Short session design

Offer 15, 25 and 45-minute modes that select a realistic slice of work and end with saved evidence.

Progress narrative

Use weekly field notes: what improved, what remains fragile and what the learner can now demonstrate.

Relevant cases

Turn sector knowledge into recurring mini-cases using banking, healthcare, retail and SaaS decisions.

Human feedback

Add mentor or peer review for portfolio work with a structured rubric and revision history.

Return trigger

Send optional reminders only when a review is due, feedback arrives or a saved goal is at risk.

What is missing

These gaps are ordered by their effect on learning outcomes, differentiation and operational readiness.

Priority	Capability	Why it matters
P0	Runnable SQL/Python labs	Transforms passive reading into observable skill evidence.
P0	Deep authored lesson content	Most non-SQL topics still need expert examples, misconceptions and technical question banks.
P0	Production identity and database	Required for reliable multi-device accounts and public launch.
P1	Public proof profile	Turns the ledger into a shareable career asset with privacy controls.
P1	Project review workflow	Adds feedback, revision and credibility beyond self-declaration.
P1	Adaptive review scheduling	Uses attempt recency and confidence to decide what should return.
P1	Analytics and experiments	Measures activation, learning success and retention without guessing.
P2	Employer challenge library	Connects skill evidence to realistic hiring scenarios.
P2	Mobile/offline experience	Supports learners with inconsistent connectivity and commute study time.
P2	Community and mentorship	Provides examples, accountability and human judgement.

Recommended delivery roadmap

Sequence the work so each phase creates evidence for the next investment decision.

0-90 days / Activation and quality

- Author the top five role paths deeply
- Build SQL sandbox prototype
- Instrument onboarding and first-session funnel
- Improve mission-based dashboard and session modes
- Launch structured project rubric

3-6 months / Evidence and retention

- Python notebook execution
- Adaptive review scheduling
- Public proof profile with privacy controls
- Mentor/peer project review beta
- Generic account recovery and production operations

6-12 months / Distribution and revenue

- University cohort dashboard
- Employer-authored challenges
- Consumer Plus pilots for labs and review
- Regional practitioner content partnerships
- Outcome study and readiness validation

Decision gates

- Do learners finish a first evidence-backed topic within their first session?
- Do users return because a clear task or review is waiting, rather than because of generic reminders?
- Does project evidence predict technical interview performance better than self-assessment?
- Will institutions pay for cohort visibility without compromising learner ownership and privacy?

Success measures and next decision

The business should measure progress toward credible learner outcomes before optimising revenue.

Activation

Profile completed; first mission opened; first evidence item saved.

Learning

Topic quiz improvement; review recovery; level and skill pass rates.

Practice

Runnable exercises completed; tests passed; debugging attempts.

Evidence

Portfolio milestones, reviewed projects and proof profiles shared.

Retention

Meaningful sessions per week and due reviews completed; avoid raw time-on-site as the main goal.

Career

Interview confidence, applications with proof links, interviews and learner-reported role outcomes.

The next investment

Build one excellent end-to-end path before expanding breadth: Data Analyst in Banking, with authored lessons, executable SQL, a sector case, technical assessments, a reviewed portfolio artifact and a public proof page. This vertical slice will test the complete value proposition and reveal which features create real learner outcomes.

Reference basis

Current DataPath source and validation records; official public product pages for DataCamp, Codecademy, Coursera, roadmap.sh, Kaggle Learn, Interview Query, Mode, StrataScratch, Maven Analytics and Dataquest; WEF Future of Jobs 2025; Microsoft career-path guidance. Links and comparison notes are maintained in BENCHMARK.md in the project.